

Manual for DSBDA Lab

Assignment 13: A Simple Scala Program.

To run a Scala program on Ubuntu, you need Java Development Kit (JDK) and the Scala installation (which includes the compiler and runtime). You may also optionally use a build tool like sbt or Coursier for managing projects and dependencies.

Required Components:

- **Java Development Kit (JDK):** Scala runs on the Java Virtual Machine (JVM), so Java is a fundamental prerequisite.

- **Recommended Version:** Scala 2.12 requires Java 8, while newer versions like Scala 3.8+ require JDK 17+. You should check which Java version is appropriate for your target Scala version.

- **Scala Installation:** This provides the scalac compiler and the scala runtime environment (which includes the interactive REPL).

- **A text editor or IDE:** A simple text editor like Vim or a dedicated IDE such as IntelliJ IDEA with the Scala plugin or VS Code with the Metals extension is needed to write your code.

General Steps to Get Started :

1. **Install the JDK:** First, ensure Java is installed on your Ubuntu system using a package manager or SDKMAN!.

2. **Install Scala:** Use a tool like [Coursier](#) or a package manager (sudo apt-get install scala) to install the Scala command-line tools.

3. **Write and Run your program:**

- **For simple scripts:** Write your code in a .scala file (e.g., HelloWorld.scala) and run it directly using the scala command (e.g., scala HelloWorld.scala).

- **For larger projects:** Use a build tool like [sbt](#) to manage your project and run your code (e.g., sbt run inside the project directory).

Follow Steps : (To Install Scala and Run a Program)

1. **Check Java Version:** `pict@pict-OptiPlex-3046:~$ java -version`

2. **Check Java Compiler:** `pict@pict-OptiPlex-3046:~$ javac -version`

3. **List all installed JDK versions:**

```
pict@pict-OptiPlex-3046:~$ sudo update-alternatives --config java
```

- [sudo] password for pict:
- There are 2 choices for the alternative java (providing /usr/bin/java).
-
- | Selection | Path | Priority | Status |
|-----------|--|----------|-------------|
| ----- | | | |
| * 0 | /usr/lib/jvm/java-11-openjdk-amd64/bin/java | 1111 | auto mode |
| 1 | /usr/lib/jvm/java-11-openjdk-amd64/bin/java | 1111 | manual mode |
| 2 | /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/java | 1081 | manual mode |
-
- Press <enter> to keep the current choice[*], or type selection number: 2
- update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/java to provide /usr/bin/java (java) in manual mode
- pict@pict-OptiPlex-3046:~\$

Reading package lists... Done

Reading state information... Done

libhwtini-runtime-java libjansi-java libjansi-native-java libjline2-java

Suggested packages:

The following NEW packages will be installed:

```
scala scala-library scala-parser-combinators scala-xml
```

0 upgraded, 8 newly installed, 0 to remove and 35 not upgraded.

Need to get 25.1 MB of archives.

After this operation, 28.6 MB of additional disk space will be used.

Do you want to continue? [Y/n] y

Get:1 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libhawtjni-runtime-java all 1.17-1 [28.8 kB]

Get:2 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libjansi-native-java all 1.8-1 [23.8 kB]

Get:3 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libjansi-java all 1.18-1 [56.8 kB]

Get:4 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libjline2-java all 2.14.6-4 [150 kB]

Get:5 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 scala-library all 2.11.12-5 [9,586 kB]

Get:6 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 scala-parser-combinators all 1.0.3-3.1 [365 kB]

Get:7 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 scala-xml all 1.0.3-3.1 [615 kB]

Get:8 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 scala all 2.11.12-5 [14.2 MB]

Fetches 25.1 MB in 23s (1,095 kB/s)

Selecting previously unselected package libhawtjni-runtime-java.

(Reading database ... 251687 files and directories currently installed.)

Preparing to unpack .../0-libhawtjni-runtime-java_1.17-1_all.deb ...

Unpacking libhawtjni-runtime-java (1.17-1) ...

Selecting previously unselected package libjansi-native-java.

Preparing to unpack .../1-libjansi-native-java_1.8-1_all.deb ...

Unpacking libjansi-native-java (1.8-1) ...

Selecting previously unselected package libjansi-java.

Preparing to unpack .../2-libjansi-java_1.18-1_all.deb ...

Unpacking libjansi-java (1.18-1) ...

Selecting previously unselected package libjline2-java.

Preparing to unpack .../3-libjline2-java_2.14.6-4_all.deb ...

Unpacking libjline2-java (2.14.6-4) ...

Selecting previously unselected package scala-library.

Preparing to unpack .../4-scala-library_2.11.12-5_all.deb ...

Unpacking scala-library (2.11.12-5) ...

Selecting previously unselected package scala-parser-combinators.

Preparing to unpack .../5-scala-parser-combinators_1.0.3-3.1_all.deb ...

Unpacking scala-parser-combinators (1.0.3-3.1) ...

Selecting previously unselected package scala-xml.

Preparing to unpack .../6-scala-xml_1.0.3-3.1_all.deb ...

Unpacking scala-xml (1.0.3-3.1) ...

Selecting previously unselected package scala.

Preparing to unpack .../7-scala_2.11.12-5_all.deb ...

Unpacking scala (2.11.12-5) ...

Setting up scala-library (2.11.12-5) ...

Setting up scala-xml (1.0.3-3.1) ...

Setting up scala-parser-combinators (1.0.3-3.1) ...

Setting up libhawtjni-runtime-java (1.17-1) ...

Setting up libjansi-native-java (1.8-1) ...

Setting up libjansi-java (1.18-1) ...

Setting up libjline2-java (2.14.6-4) ...

Setting up scala (2.11.12-5) ...

update-alternatives: using /usr/share/scala-2.11/bin/scala to provide
/usr/bin/scala (scala) in auto mode

5. Type a Hello World Program : pict@pict-OptiPlex-3046:~\$ gedit hello.scala

```
object HelloWorld {  
  def main(args: Array[String]): Unit = {  
    println("Hello, World!")  
  }  
}
```

6. Compile a Program : pict@pict-OptiPlex-3046:~\$ scalac hello.scala

7. Run a Program : pict@pict-OptiPlex-3046:~\$ scala hello.scala

Hello, World!

pict@pict-OptiPlex-3046:~\$

8. Type a Factorial Program :

pict@pict-OptiPlex-3046:~\$ gedit Factorial.scala

```
object FactorialCalculator {  
  def factorial(n: Int): Int = {  
    if (n == 0)  
      1 // Base case  
    else  
      n * factorial(n - 1) // Recursive case  
  }  
  
  def main(args: Array[String]): Unit = {  
    val number = 5  
    println(s"Factorial of $number: ${factorial(number)}")  
  }  
}
```

9. Compile a Program :

pict@pict-OptiPlex-3046:~\$ scalac Factorial.scala

10. Run a Program :

pict@pict-OptiPlex-3046:~\$ scala Factorial.scala

Factorial of 5: 120

pict@pict-OptiPlex-3046:~\$